

Final Exam

Can monthly fuel purchases for U.S. power plants be grouped into distinct segments based on fuel quality, cost, and quantity, and what do these segments reveal about power generation practices?

ALL files relating to this assignment can be found at https://github.com/Dylan-Hamilton3412/64060_-Dhamil28

Import required libraries for data manipulation and cluster analysis
`library(tidyverse)`

Warning: package 'tidyverse' was built under R version 4.5.3

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'forcats' was built under R version 4.5.3

Warning: package 'lubridate' was built under R version 4.5.2

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr 1.1.4 ✓ readr 2.1.5

✓ forcats 1.0.1 ✓ stringr 1.5.2

✓ ggplot2 4.0.0 ✓ tibble 3.3.0

✓ lubridate 1.9.4 ✓ tidyr 1.3.1

✓ purrr 1.2.0

— Conflicts — tidyverse_conflicts() —

✗ dplyr::filter() masks stats::filter()

✗ dplyr::lag() masks stats::lag()

i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

`library(dplyr)`

`library(factoextra)`

Warning: package 'factoextra' was built under R version 4.5.3

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanova.com/en/product/practical-guide-to-principal-component-methods-in-r/>

`library(ISLR)`

`library(readr)`

`library(caret)`

```

## Warning: package 'caret' was built under R version 4.5.2

## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift

## Set seed to ensure reproducibility of the K-means results
set.seed(1234)

## import the data set
FuelData <- read.csv("out_eia923__monthly_fuel_receipts_costs.csv", stringsAs
Factors = FALSE)
summary(FuelData)

##  report_date      plant_id_eia  plant_id_pudl  plant_name_eia
## Length:296604    Min.      :    3  Min.      :    1  Length:296604
## Class :character  1st Qu.: 2399  1st Qu.:  445  Class :character
## Mode  :character  Median : 6664  Median : 1852  Mode  :character
##                      Mean   :21586  Mean    : 2275
##                      3rd Qu.:55051  3rd Qu.: 3761
##                      Max.    :67766  Max.    :19126
##  utility_id_eia  utility_id_pudl  utility_name_eia      state
## Min.      :    7  Min.      :    7  Length:296604  Length:296604
## 1st Qu.: 7140  1st Qu.:  246  Class :character  Class :character
## Median :14354  Median : 1178  Mode  :character  Mode  :character
## Mean   :20281  Mean   : 1660
## 3rd Qu.:20860  3rd Qu.: 2689
## Max.   :67826  Max.   :17784
##  fuel_type_code_pudl  fuel_received_units  fuel_mmbtu_per_unit
## Length:296604      Min.      :    1  Min.      :  0.000
## Class :character    1st Qu.: 10092  1st Qu.:  1.025
## Mode  :character    Median : 114322  Median :  1.046
##                      Mean   :  675225  Mean   :  6.239
##                      3rd Qu.: 697712  3rd Qu.:  5.900
##                      Max.    :48206176  Max.    :1049.000
##  fuel_cost_per_mmbtu  fuel_received_mmbtu  total_fuel_cost
## Length:296604      Min.      :    0  Length:296604
## Class :character    1st Qu.:  22214  Class :character
## Mode  :character    Median : 384482  Mode  :character
##                      Mean   : 1426788
##                      3rd Qu.: 1950661
##                      Max.    :24881922
##  fuel_cost_per_mmbtu_source  sulfur_content_pct  ash_content_pct
## Length:296604              Min.      :0.0000  Min.      : 0.000
## Class :character            1st Qu.:0.0000  1st Qu.: 0.000
## Mode  :character            Median :0.0000  Median : 0.000

```

```
##               Mean    :0.3015      Mean    : 2.136
##               3rd Qu.:0.2100      3rd Qu.: 0.000
##               Max.    :6.8000      Max.    :73.979
## mercury_content_ppm moisture_content_pct chlorine_content_ppm
## Length:296604      Length:296604      Length:296604
## Class :character    Class :character    Class :character
## Mode  :character    Mode  :character    Mode  :character
##
##
##
## data_maturity
## Length:296604
## Class :character
## Mode  :character
##
##
##
```

str(FuelData)

```
## 'data.frame':    296604 obs. of  21 variables:
## $ report_date      : chr  "2008-01-01T00:00:00.000Z" "2008-02-01
T00:00:00.000Z" "2008-03-01T00:00:00.000Z" "2008-04-01T00:00:00.000Z" ...
## $ plant_id_eia     : int   3 3 3 3 3 3 3 3 3 3 ...
## $ plant_id_pudl    : int   32 32 32 32 32 32 32 32 32 32 ...
## $ plant_name_eia   : chr   "Barry" "Barry" "Barry" "Barry" ...
## $ utility_id_eia   : int   195 195 195 195 195 195 195 195 195 19
5 ...
## $ utility_id_pudl  : int   18 18 18 18 18 18 18 18 18 18 ...
## $ utility_name_eia : chr   "Alabama Power Co" "Alabama Power Co"
"Alabama Power Co" "Alabama Power Co" ...
## $ state            : chr   "AL" "AL" "AL" "AL" ...
## $ fuel_type_code_pudl : chr   "coal" "coal" "coal" "coal" ...
## $ fuel_received_units : int   311653 246983 292376 353083 337140 299
531 282146 283287 414388 372175 ...
## $ fuel_mmbtu_per_unit : num   23 23 23 22.9 22.9 ...
## $ fuel_cost_per_mmbtu : chr   "2.131683826446533" "2.143523454666137
7" "2.57438325881958" "2.787388324737549" ...
## $ fuel_received_mmbtu : num   7183512 5679396 6720962 8092480 771589
2 ...
## $ total_fuel_cost   : chr   "15312976" "12173918" "17302334" "2255
6884" ...
## $ fuel_cost_per_mmbtu_source: chr   "mixed" "mixed" "mixed" "mixed" ...
## $ sulfur_content_pct : num   0.488 0.502 0.506 0.5 0.528 ...
## $ ash_content_pct   : num   5.45 5.59 5.51 5.59 5.31 ...
## $ mercury_content_ppm : chr   "null" "null" "null" "null" ...
## $ moisture_content_pct : chr   "null" "null" "null" "null" ...
## $ chlorine_content_ppm : chr   "null" "null" "null" "null" ...
## $ data_maturity     : chr   "final" "final" "final" "final" ...
```

```

FuelData[FuelData == "null"] <- NA

## Remove variables with more than 40% missing values
Missing_Percent <- colSums(is.na(FuelData)) / nrow(FuelData) * 100
Missing_Percent

##          report_date          plant_id_eia
##          0.000000          0.000000
##          plant_id_pudl          plant_name_eia
##          0.000000          0.000000
##          utility_id_eia          utility_id_pudl
##          0.000000          0.000000
##          utility_name_eia          state
##          0.000000          0.000000
##          fuel_type_code_pudl          fuel_received_units
##          0.000000          0.000000
##          fuel_mmbtu_per_unit          fuel_cost_per_mmbtu
##          0.000000          9.674178
##          fuel_received_mmbtu          total_fuel_cost
##          0.000000          9.673504
##          fuel_cost_per_mmbtu_source          sulfur_content_pct
##          9.664401          0.000000
##          ash_content_pct          mercury_content_ppm
##          0.000000          37.251689
##          moisture_content_pct          chlorine_content_ppm
##          88.053769          88.053769
##          data_maturity
##          0.000000

FuelData <- FuelData[, Missing_Percent < 33]
FuelData <- na.omit(FuelData)

## Convert categorical variables to factors
FuelData$state <- as.factor(FuelData$state)
FuelData$fuel_type_code_pudl <- as.factor(FuelData$fuel_type_code_pudl)
FuelData$plant_name_eia <- as.factor(FuelData$plant_name_eia)
FuelData$utility_name_eia <- as.factor(FuelData$utility_name_eia)

## Convert Categorical to numeric
FuelData$fuel_cost_per_mmbtu <- as.numeric(FuelData$fuel_cost_per_mmbtu)
FuelData$total_fuel_cost <- as.numeric(FuelData$total_fuel_cost)

## Convert numeric to categorical
FuelData$plant_id_eia <- as.character(FuelData$plant_id_eia)
FuelData$plant_id_pudl <- as.character(FuelData$plant_id_pudl)
FuelData$utility_id_eia <- as.character(FuelData$utility_id_eia)
FuelData$utility_id_pudl <- as.character(FuelData$utility_id_pudl)
summary(FuelData)

```

```

## report_date      plant_id_eia      plant_id_pudl
## Length:267910    Length:267910    Length:267910
## Class :character  Class :character  Class :character
## Mode :character   Mode :character   Mode :character
##
##
##
##
##                plant_name_eia      utility_id_eia      utility_id_pudl
##
## Walter Scott Jr Energy Center:    581    Length:267910    Length:267910
##
## Victor J Daniel Jr                :    569    Class :character  Class :character
## Williams                          :    548    Mode :character   Mode :character
## Clay Boswell                      :    547
##
## Georgia-Pacific Cedar Springs:    534
##
## Marion                            :    527
##
## (Other)                           :264604
##
##                utility_name_eia      state      fuel_type_code_pudl
##
## Tennessee Valley Authority :    5153    TX      : 27973    coal: 60228
##
## Virginia Electric & Power Co :    4967    FL      : 13647    gas :166598
##
## PacifiCorp                    :    3367    CA      : 13571    oil : 41084
##
## Duke Energy Carolinas, LLC    :    3253    PA      : 13154
##
## Georgia Power Co              :    3087    IL      : 10977
##
## Public Service Co of Colorado:    2911    NY      : 9713
##
## (Other)                       :245172    (Other):178875
##
## fuel_received_units fuel_mmbtu_per_unit fuel_cost_per_mmbtu
## Min. :      1    Min. : 0.078    Min. : -71.93
## 1st Qu.: 11149    1st Qu.: 1.024    1st Qu.: 2.62
## Median : 120622    Median : 1.043    Median : 3.97
## Mean : 691810     Mean : 6.118     Mean : 15.87
## 3rd Qu.: 721398    3rd Qu.: 5.880    3rd Qu.: 6.97
## Max. :48206176     Max. :1049.000    Max. :562572.19
##

```

```
## fuel_received_mmbtu total_fuel_cost fuel_cost_per_mmbtu_source
## Min. :8.000e-01 Min. : -3123662 Length:267910
## 1st Qu.:2.298e+04 1st Qu.: 174406 Class :character
## Median :3.984e+05 Median : 1418210 Mode :character
## Mean :1.453e+06 Mean : 4635326
## 3rd Qu.:2.008e+06 3rd Qu.: 6273863
## Max. :2.488e+07 Max. :656044928
##
## sulfur_content_pct ash_content_pct data_maturity
## Min. :0.0000 Min. : 0.000 Length:267910
## 1st Qu.:0.0000 1st Qu.: 0.000 Class :character
## Median :0.0000 Median : 0.000 Mode :character
## Mean :0.2928 Mean : 2.075
## 3rd Qu.:0.2000 3rd Qu.: 0.000
## Max. :6.8000 Max. :73.979
##
```

```
str(FuelData)
```

```
## 'data.frame': 267910 obs. of 18 variables:
## $ report_date : chr "2008-01-01T00:00:00.000Z" "2008-02-01
T00:00:00.000Z" "2008-03-01T00:00:00.000Z" "2008-04-01T00:00:00.000Z" ...
## $ plant_id_eia : chr "3" "3" "3" "3" ...
## $ plant_id_pudl : chr "32" "32" "32" "32" ...
## $ plant_name_eia : Factor w/ 1717 levels "2 AC Station",...: 10
4 104 104 104 104 104 104 104 104 ...
## $ utility_id_eia : chr "195" "195" "195" "195" ...
## $ utility_id_pudl : chr "18" "18" "18" "18" ...
## $ utility_name_eia : Factor w/ 1045 levels "Acadia Power Partner
s",...: 28 28 28 28 28 28 28 28 28 28 ...
## $ state : Factor w/ 50 levels "AK","AL","AR",...: 2 2
2 2 2 2 2 2 2 2 ...
## $ fuel_type_code_pudl : Factor w/ 3 levels "coal","gas","oil": 1 1
1 1 1 1 1 1 1 1 ...
## $ fuel_received_units : int 311653 246983 292376 353083 337140 299
531 282146 283287 414388 372175 ...
## $ fuel_mmbtu_per_unit : num 23 23 23 22.9 22.9 ...
## $ fuel_cost_per_mmbtu : num 2.13 2.14 2.57 2.79 2.79 ...
## $ fuel_received_mmbtu : num 7183512 5679396 6720962 8092480 771589
2 ...
## $ total_fuel_cost : num 15312976 12173918 17302334 22556884 21
512616 ...
## $ fuel_cost_per_mmbtu_source: chr "mixed" "mixed" "mixed" "mixed" ...
## $ sulfur_content_pct : num 0.488 0.502 0.506 0.5 0.528 ...
## $ ash_content_pct : num 5.45 5.59 5.51 5.59 5.31 ...
## $ data_maturity : chr "final" "final" "final" "final" ...
## - attr(*, "na.action")= 'omit' Named int [1:28694] 9614 9615 9616 9617 96
18 9619 9620 9621 9625 9626 ...
## ... attr(*, "names")= chr [1:28694] "9614" "9615" "9616" "9617" ...
```

```

## Count negative fuel costs
sum(FuelData$total_fuel_cost < 0, na.rm = TRUE)

## [1] 43

## View the negative records
Negative_Costs <- FuelData %>%
  filter(total_fuel_cost < 0)

## Summary of negative values
summary(Negative_Costs$total_fuel_cost)

##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## -3123662 -1326836  -313048  -772605  -103043   -2365

## Randomly sample approximately 2% of the observations
Sample_Index <- sample(1:nrow(FuelData), 0.02 * nrow(FuelData))
FuelSample <- FuelData[Sample_Index,]

## Create a dataset containing only the numeric variables for clustering
Cluster_Data <- FuelSample %>%
  select(fuel_received_units,
         fuel_mmbtu_per_unit,
         fuel_cost_per_mmbtu,
         fuel_received_mmbtu,
         total_fuel_cost,
         sulfur_content_pct,
         ash_content_pct)

## Standardize the variables so each contributes equally
Cluster_Data <- scale(Cluster_Data)
Cluster_Data <- as.data.frame(Cluster_Data)
summary(Cluster_Data)

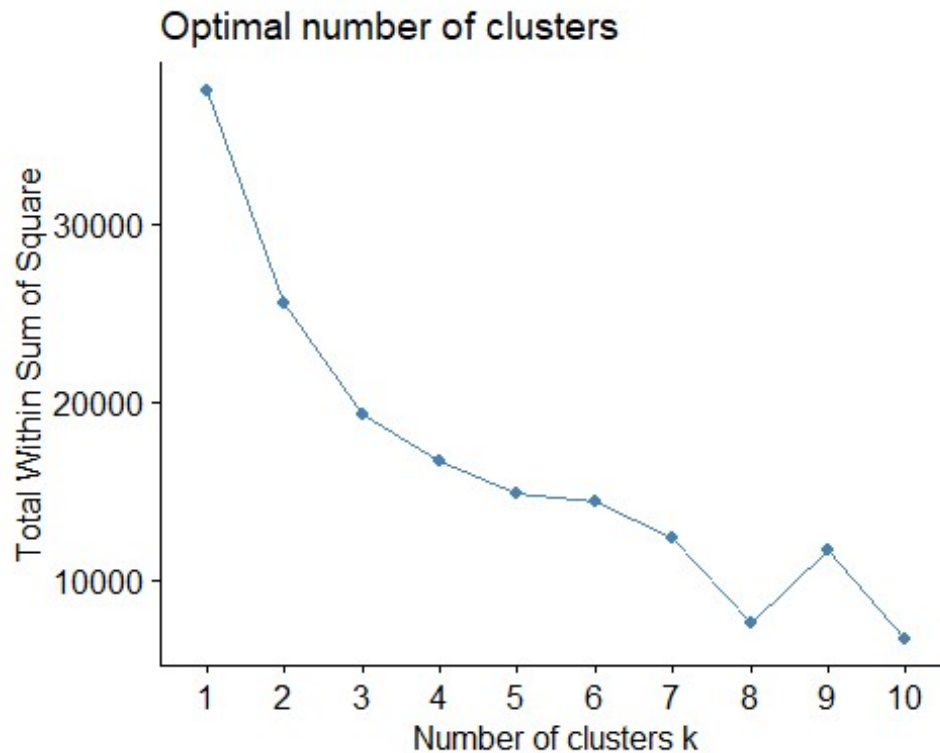
## fuel_received_units fuel_mmbtu_per_unit fuel_cost_per_mmbtu
## Min.      :-0.541218  Min.      :-0.74661   Min.      :-0.20658
## 1st Qu.: -0.533237   1st Qu.: -0.63210   1st Qu.: -0.11829
## Median : -0.447844   Median : -0.62978   Median : -0.08700
## Mean    : 0.000000    Mean    : 0.00000    Mean    : 0.00000
## 3rd Qu.: 0.003686    3rd Qu.: -0.03901   3rd Qu.: -0.01852
## Max.    : 9.799798    Max.    : 2.73347    Max.    : 39.49235
## fuel_received_mmbtu total_fuel_cost  sulfur_content_pct ash_content_pct
## Min.      :-0.6197    Min.      :-0.6044    Min.      :-0.4019    Min.      :-0.4006
## 1st Qu.: -0.6105     1st Qu.: -0.5822     1st Qu.: -0.4019     1st Qu.: -0.4006
## Median : -0.4562     Median : -0.4224     Median : -0.4019     Median : -0.4006

```

```
## Mean : 0.0000 Mean : 0.0000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 0.2622 3rd Qu.: 0.2191 3rd Qu.: -0.1284 3rd Qu.: -0.4006
## Max. : 8.8334 Max. : 16.7015 Max. : 7.7894 Max. : 11.7537
```

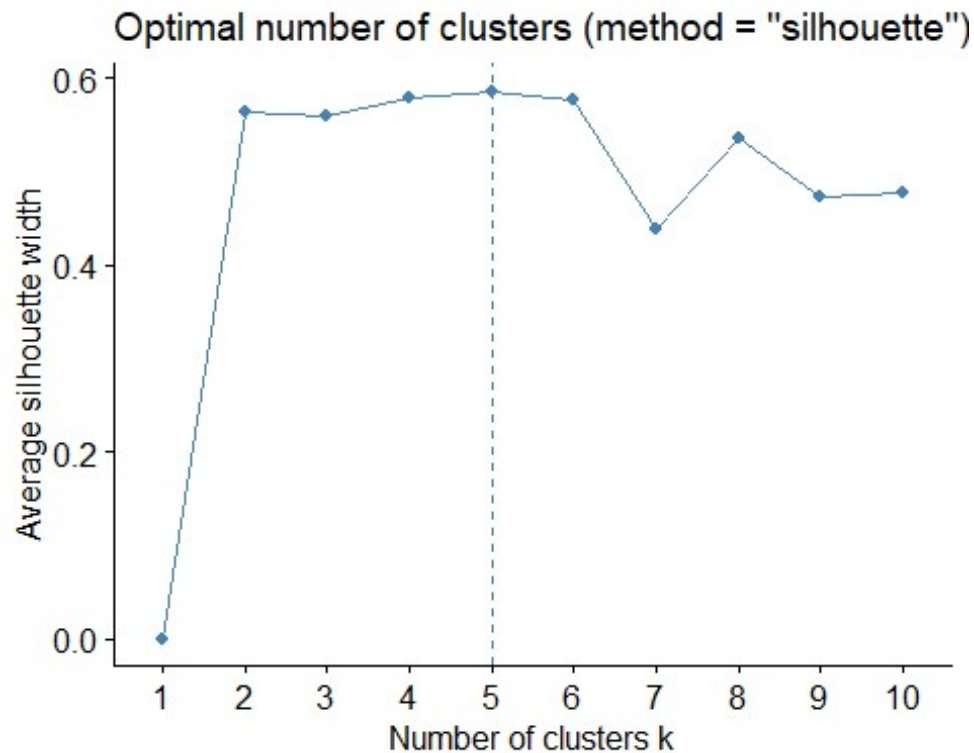
Run an elbow chart to determine the best fit of K without over-fitting the algorithm

```
fviz_nbclust(Cluster_Data, kmeans, method = "wss")
```



Confirm the choice of K using silhouette values to maximize cluster separation

```
fviz_nbclust(Cluster_Data, kmeans, method = "silhouette")
```

##K Value will be 5 for the remainder of this project

Based on both methods, K = appears to provide a reasonable balance between cluster separation and model complexity

```
Num_KClust <- kmeans(Cluster_Data, centers = 5, nstart = 25)
```

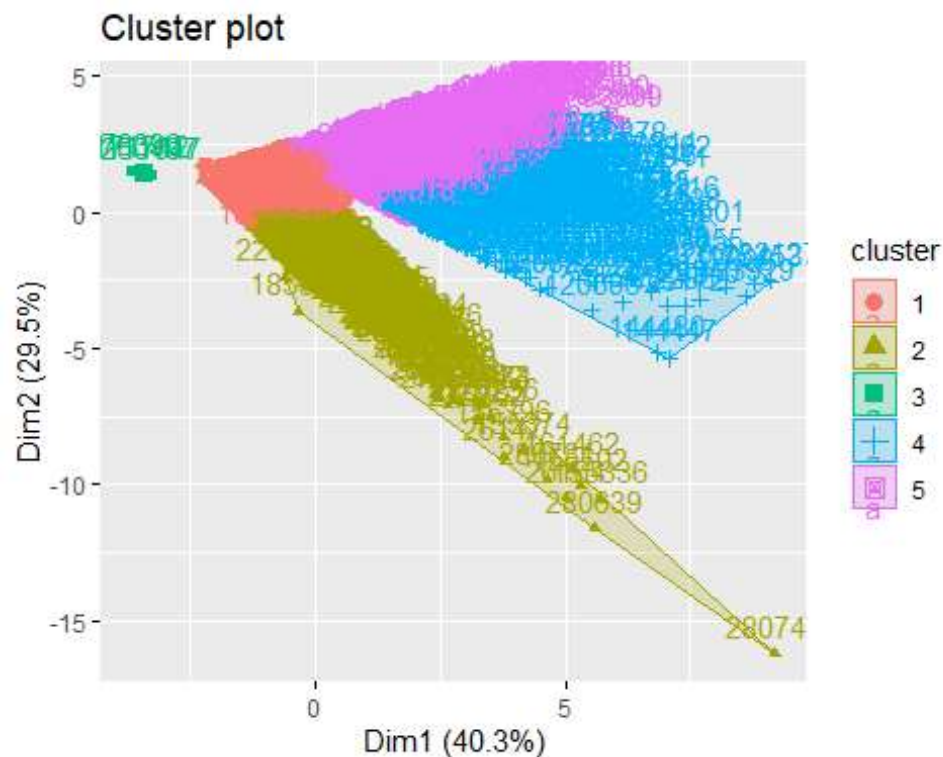
Display the number of transactions assigned to each cluster

```
Num_KClust$size
```

```
## [1] 3377 752 4 273 952
```

Visualize the clusters using a plot

```
fviz_cluster(Num_KClust, data = Cluster_Data)
```



Append cluster assignments to both the scaled numerical data and the original dataset

```
Cluster_Data <- cbind(Cluster_Data, "Cluster" = Num_KClust$cluster)
FuelSample <- cbind(FuelSample, "Cluster" = Num_KClust$cluster)
```

Calculate average values for each variable within each cluster to aid interpretation

```
Cluster_Means <- aggregate(FuelSample[, c(10:14, 16:17)], by = list(Cluster = FuelSample$Cluster), mean)
Cluster_Means
```

```
##   Cluster fuel_received_units fuel_mmbtu_per_unit fuel_cost_per_mmbtu
## 1      1      295009.29      2.165752      8.681774
## 2      2     3319236.11      1.027542      4.425882
## 3      3       177.75      1.004750     1566.011322
## 4      4     432668.94     21.002320      2.337193
## 5      5     104123.39     20.369977      2.417996
##   fuel_received_mmbtu total_fuel_cost sulfur_content_pct ash_content_pct
## 1      309092.426     1352732.4      0.03247256      0.000000
## 2     3400167.550     14458331.9      0.00000000      0.000000
## 3       178.554      288688.6      0.00000000      0.000000
## 4     8770355.716     20234505.6      1.50776776      8.929485
## 5     1982933.171      4432248.1      1.10667784      9.292875
```

```
Std_Dev_Means <- aggregate(Cluster_Data, by = list(Cluster = Cluster_Data$Cluster), mean)
```

Export cluster means for further examination. File added to GitHub along with assignment

```
write.csv(Cluster_Means, "Cluster_Means.csv", row.names = FALSE)
write.csv(Std_Dev_Means, "Std_Dev_Means.csv", row.names = FALSE)
```

Compare cluster averages against overall dataset averages to identify distinguishing characteristics

```
summary(FuelSample)
```

```
## report_date      plant_id_eia      plant_id_pudl
## Length:5358      Length:5358      Length:5358
## Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character
##
##
##
##
##              plant_name_eia  utility_id_eia      utility_id_pudl
##
## Bowen                      : 21  Length:5358      Length:5358
##
## E W Brown                  : 15  Class :character  Class :character
##
## F B Culley                  : 14  Mode  :character  Mode  :character
##
## Scrubgrass Generating Company LP: 14
##
## South Oak Creek            : 14
##
## Wateree                    : 14
##
## (Other)                    :5266
##
##              utility_name_eia      state      fuel_type_code_pudl
## Tennessee Valley Authority : 105  TX      : 553  coal:1225
## Virginia Electric & Power Co : 94   CA      : 269  gas :3333
## Georgia Power Co           : 79   FL      : 256  oil : 800
## Public Service Co of Colorado: 70   IL      : 246
## Duke Energy Carolinas, LLC  : 66   PA      : 243
## Wisconsin Electric Power Co : 65   GA      : 204
## (Other)                     :4879  (Other):3587
## fuel_received_units fuel_mmbtu_per_unit fuel_cost_per_mmbtu
## Min. : 1 Min. : 0.08644 Min. : -1.245
## 1st Qu.: 10210 1st Qu.: 1.02400 1st Qu.: 2.626
## Median : 119448 Median : 1.04300 Median : 3.997
## Mean : 692340 Mean : 6.19939 Mean : 7.811
## 3rd Qu.: 697055 3rd Qu.: 5.88000 3rd Qu.: 6.999
## Max. :13228464 Max. :28.58000 Max. :1739.020
```

```
##
## fuel_received_mmbtu total_fuel_cost fuel_cost_per_mmbtu_source
## Min. : 2 Min. : -2365 Length:5358
## 1st Qu.: 21793 1st Qu.: 170454 Class :character
## Median : 388107 Median : 1413816 Mode :character
## Mean : 1471219 Mean : 4700545
## 3rd Qu.: 2093680 3rd Qu.: 6405471
## Max. :22441826 Max. :134657440
##
## sulfur_content_pct ash_content_pct data_maturity Cluster
## Min. :0.0000 Min. : 0.000 Length:5358 Min. :1.000
## 1st Qu.:0.0000 1st Qu.: 0.000 Class :character 1st Qu.:1.000
## Median :0.0000 Median : 0.000 Mode :character Median :1.000
## Mean :0.2939 Mean : 2.106 Mean :2.005
## 3rd Qu.:0.2000 3rd Qu.: 0.000 3rd Qu.:2.000
## Max. :5.9900 Max. :63.900 Max. :5.000
##
```

compare Fuel type category to identifies any similarities in the clusters
table(FuelSample\$Cluster, FuelSample\$fuel_type_code_pudl)

```
##
## coal gas oil
## 1 0 2577 800
## 2 0 752 0
## 3 0 4 0
## 4 273 0 0
## 5 952 0 0
```

Even without the fuel type being included in the clustering there is a strong separation when tabling showing a clear pattern between the clustering variables and the fuel type

Cluster 1 – Low-Volume Natural Gas and Oil Plants -Below-average fuel received and energy production -Below-average total fuel cost -Low sulfur and ash content -Contains the majority of natural gas plants and all oil plants -Interpretation: Smaller or lower-utilization clean-burning facilities, including many peaking plants.

Cluster 2 – Large-Scale Natural Gas Plants -Highest fuel received units (+2.05) -Above-average total fuel received and fuel cost -Very low sulfur and ash content -Entirely natural gas -Interpretation: Large, high-output natural gas generating stations operating at a much larger scale than Cluster 1.

Cluster 3 – Fuel Cost Outlier -Extremely high fuel cost per MMBtu (+35.55) -All other variables below average -Only 4 observations -Interpretation: This cluster represents unusual fuel purchase records or pricing anomalies rather than a meaningful operational category. Because it contains only four observations, it should be interpreted as an outlier cluster rather than a typical group of power plants.

Cluster 4 – High-Output Coal Plants -Very high fuel energy content (+1.81) -Highest total energy received (+3.07) -High total fuel cost (+2.00) -Highest sulfur and ash content - Entirely coal -Interpretation: Large baseload coal plants with high fuel consumption and relatively high emissions characteristics.

Cluster 5 – Moderate Coal Plants -Above-average fuel energy content (+1.73) -Moderate fuel received and total fuel cost -Elevated sulfur and ash content -Entirely coal - Interpretation: Medium-sized coal facilities with similar fuel quality to Cluster 4 but operating at a smaller scale.

Overall statement: It appears that the predominantly coal clusters have the highest amount of MMBtu per unit meaning they output the most energy per unit of coal however at the same point it seems that these clusters also contribute to the most pollution by looking at the sulfur content and ash content.